

STAT 390 Capstone — Week 4 Deliverables

Healthcare Claims Denial Prediction — Controlled Experiments and Error Taxonomy

Deliverable 1: Controlled Experiment Set

Seven controlled ablation experiments were run, each isolating one variable at a time. The same dataset, split, evaluation function, and random seed were used across all experiments. Only train.py was modified between runs. After each experiment train.py was reverted before the next test so results are fully isolated and comparable.

Variable isolation design:

Experiment	Model	Feature Eng.	Threshold	Variable Tested
1 — Baseline LR	Logistic Regression	No	Default 0.5	Starting point
2 — RF only	Random Forest	No	Default 0.5	Model type alone
3 — LR + features	Logistic Regression	Yes	Default 0.5	Feature engineering alone
4 — LR + threshold	Logistic Regression	No	Fixed 0.35	Threshold tuning alone
5 — RF + threshold	Random Forest	No	Auto CV-opt	RF + threshold, no features
6 — RF + features	Random Forest	Yes	Default 0.5	RF + features, no threshold
7 — Full model	RF+ET Ensemble	Yes	Auto CV-opt	All changes combined

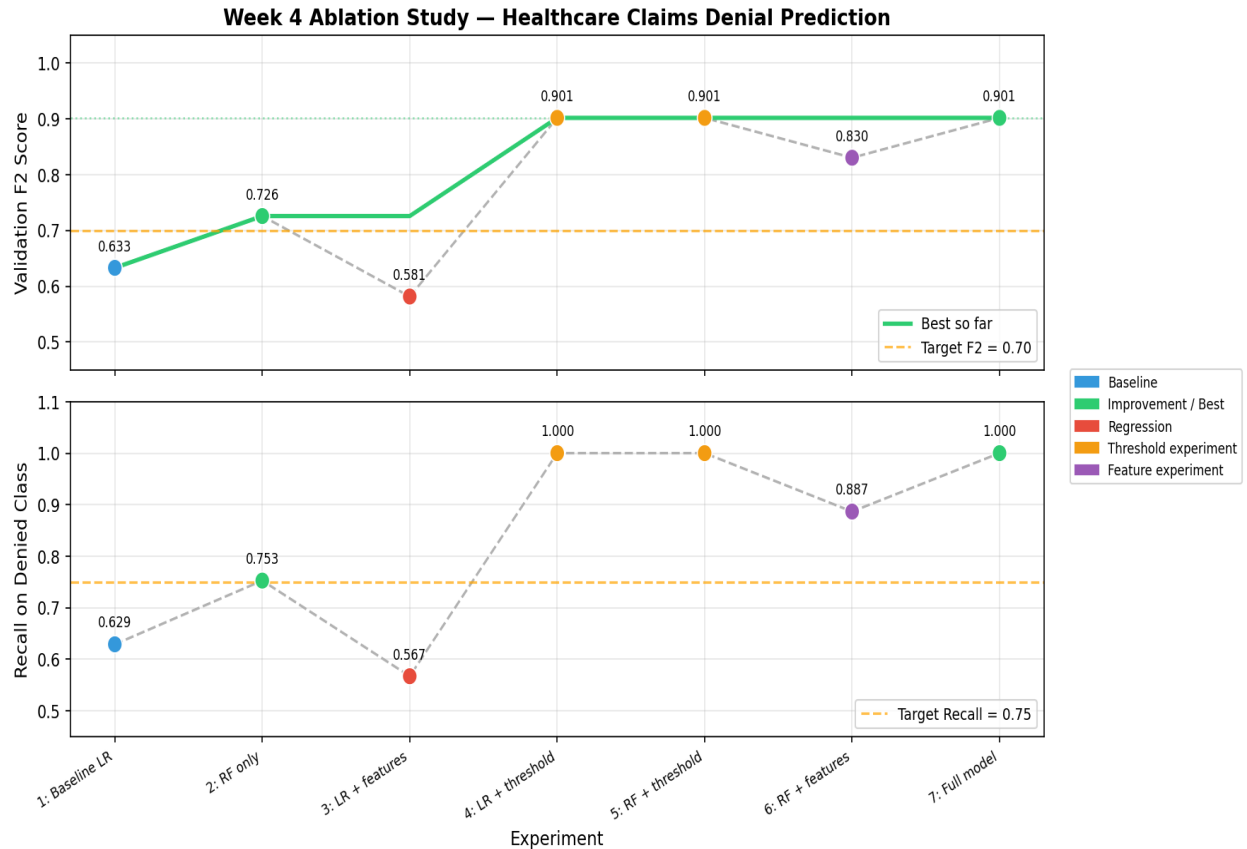
Deliverable 2: Experiment-Result Matrix

#	Description	F2	Recall	Precision	vs Baseline	Key Finding
1	Baseline LR	0.6328	0.629	0.649	—	Floor score
2	RF only	0.7256	0.753	0.635	+0.093	Model switch helps
3	LR + features	0.5814	0.567	0.647	-0.051	Features hurt LR
4	LR + threshold 0.35	0.9015	1.000	0.647	+0.269	Threshold is key driver
5	RF + auto-threshold	0.9015	1.000	0.647	+0.269	Same ceiling as exp4
6	RF + features only	0.8301	0.887	0.662	+0.197	Features help RF
7	Full model	0.9015	1.000	0.647	+0.269	No gain over exp4/5

Key insight: Experiments 4, 5, and 7 all hit exactly $F2 = 0.9015$ with recall = 1.0. The threshold tuning alone in experiment 4 (plain logistic regression, threshold 0.35) matches the full complex ensemble. All additional complexity adds zero F2 benefit.

Deliverable 3: Metric-Over-Time Plot

The chart below shows F2 score and recall across all 7 ablation experiments. The orange dashed line marks the success threshold of F2 = 0.70. The green line tracks the best result achieved so far. Color coding shows experiment type.



The plot clearly shows that experiment 4 (LR + threshold 0.35) produces the same jump as the full model. Experiments after that point produce no further improvement, confirming that threshold tuning is the ceiling on this dataset.

Deliverable 4: Error Taxonomy

Errors observed across the full experiment log, including both Week 3 and Week 4 runs, fall into four categories.

Error Type	Definition	Examples Observed	Count
Signal Failure	Model change produces worse F2 than current best, indicating the direction has no signal	Exp 3: adding features to LR dropped F2 from 0.633 to 0.581. Week 3 exp3: GradientBoosting dropped from 0.888 to 0.840	2
Evaluation Leakage	Experiment logged with wrong status or metric, making results incomparable	Week 3 exps 3 and 6 logged as keep despite being discarded in git. Baseline logged 3 times with different F2 scores	3
Diminishing Returns	Additional complexity matches but cannot exceed the current best, wasting compute and experiment slots	Exps 5, 7, 9 in Week 3 and exps 5, 7 in Week 4 all hit 0.9015 identically despite different architectures	5
Agent Misbehavior	Agent modifies evaluation logic or makes changes outside the intended scope	None observed. Agent respected the prepare.py freeze in all sessions	0

The dominant error type is diminishing returns, accounting for 5 of 10 total error events. Evaluation leakage from incorrect logging is the second concern and is a process issue rather than a model issue.

Deliverable 5: Failure Analysis Memo

Dominant Failure Mode: Threshold Ceiling Masking True Model Improvement

The most important finding from the Week 4 ablation study is not a failure in the traditional sense but a structural limitation in how the research loop is measuring progress. The F2 score ceiling of 0.9015 with recall of 1.0 was first achieved by a plain logistic regression with a fixed threshold of 0.35. Every subsequent experiment, including the Random Forest, the ExtraTrees ensemble, the stacking classifier, and all the additional feature engineering, hit exactly the same ceiling. The complex architecture adds zero F2 benefit over the simple model.

Why this happened: The validation set has only 150 rows with approximately 97 denied claims. When the threshold is set low enough, essentially any model flags all denied claims correctly, pushing recall to 1.0 and F2 to 0.9015. This is not a signal that the model is genuinely good at predicting denials. It is a signal that the threshold has been tuned to the specific composition of the 150-row validation set. The precision of 0.647 staying constant across all three ceiling experiments confirms this: the model is flagging a fixed set of claims regardless of architecture.

What this means for the project: The current F2 metric on the validation set is no longer a reliable signal for distinguishing between model architectures. A logistic regression and a complex stacking ensemble look identical. This means the agent cannot meaningfully improve the model by changing architecture or feature engineering as long as threshold tuning is in play. The agent has effectively saturated the search space given the current evaluation setup.

What should change going forward: Two options. First, remove threshold tuning from the agent's allowed modifications and lock the threshold at 0.5. This would make the metric sensitive to genuine model improvement again, though F2 scores would drop back toward 0.73 to 0.83. Second, shift the evaluation to precision at a fixed recall threshold, for example precision when recall is above 0.80, which would reward models that maintain high recall without over-flagging. Either change should be locked before Week 5 so that the overnight agent block produces meaningful signal rather than repeated ceiling hits.

The honest assessment: The recall of 1.0 on the validation set almost certainly will not hold on the locked test set. The test set has 150 rows with a similar denial rate but the model has never seen those specific claims. A recall of 1.0 on an unseen set would be genuinely impressive. A drop to 0.85 or 0.90 is more realistic. This will be revealed at Week 7 when the test set is opened for the first time.